Service Integration Testing

17-423/723 Software System Design

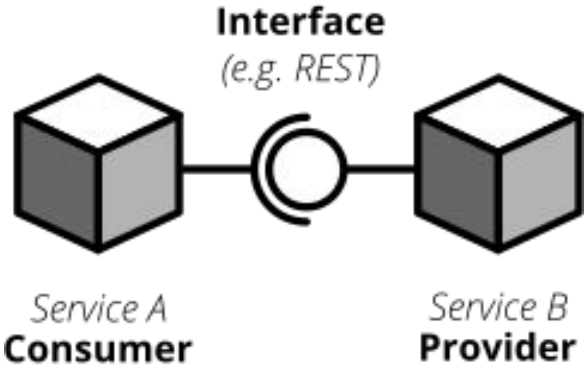
Recitation 6
Mar 13, 2026

Integration Testing

- Testing that multiple components/services work together as intended
- **Big-bang test**: Deploy & execute all services together under different test inputs & check the output
- **Q. Why is challenging to do in practice?**
  - Dependencies on external services
  - Debugging/localizing a buggy behavior to a single component
  - Generally, slow and expensive to set up
- Eventually, you will need some big-bang tests, to make sure that your system works under the production-like environment
- But can we also do something that’s less expensive & difficult?

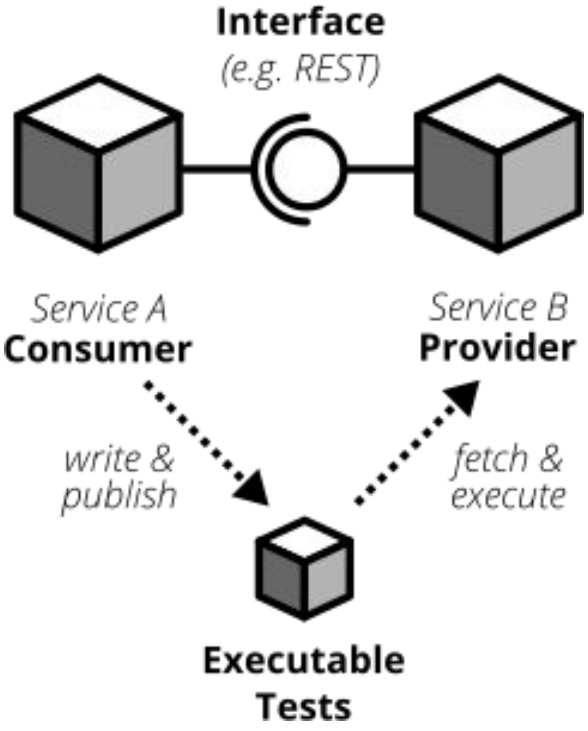
Contract Testing

- An incremental, service-by-service approach to integration testing
- **Provider:** Provides data to consumers
- **Consumer:** Processes data obtained from a provider
- **Consumer-driven Contract (CDC)**: Describes what the consumer expects from the provider as an output



Contract Testing: Steps

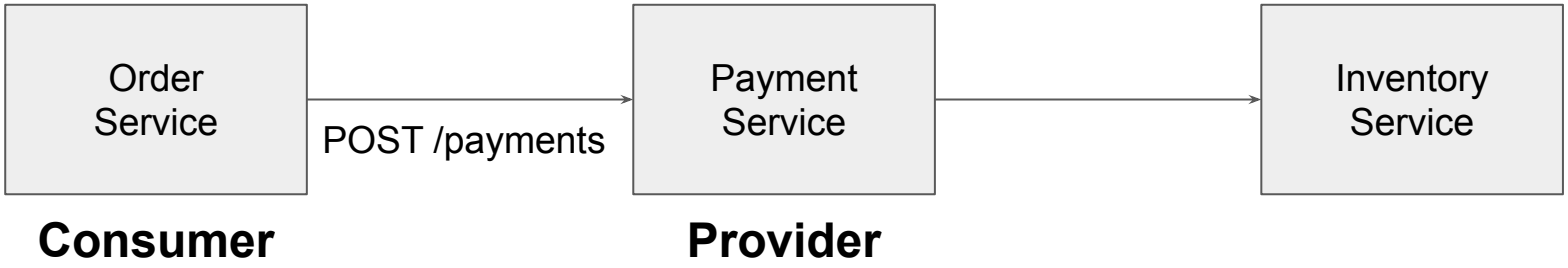
1. **Write a consumer unit test**: To do so, create a mock for the provider.
2. **Create a contract** (CDC) that describes (1) an input from the consumer to the provider and (2) the expected output.
3. **Run the consumer unit test** (with the mock) → done by the consumer
4. **Publish the contracts:** Share all CDCs in a machine-readable form.
5. **Test the provider**: Run all CDCs against the provider → done by the provider



Example: Online Store System

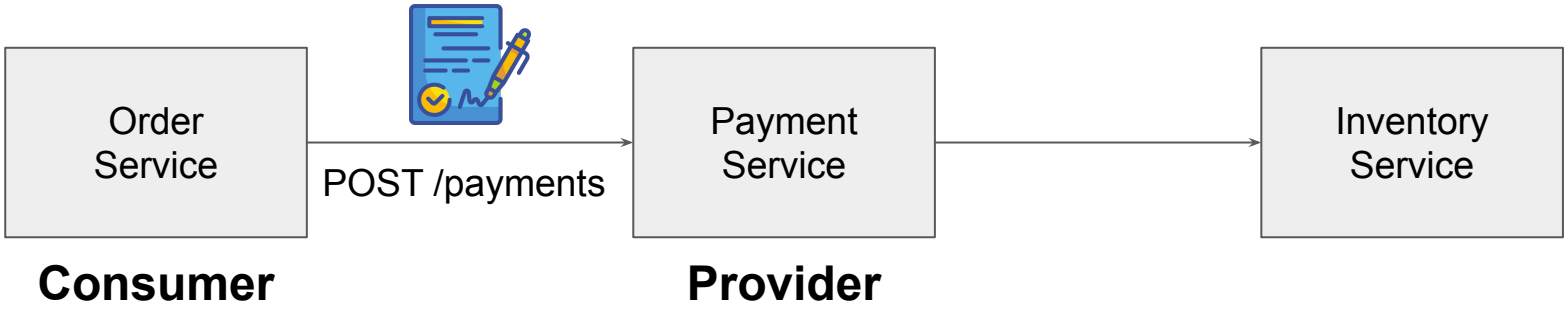
- **Order Service**: Handles orders from customers.
- **Payment Service**: Processes payments for orders.
- **Inventory Service**: Manage the stock levels for different items. Reserve items for a processed order.

Contract between Order & Payment Services



- **Consumer unit test**: Test a scenario where a customer order is successfully handled and finalized.
- The unit test includes (1) a mock of Payment Service and (2) a contract from Order to Payment Service

Contract between Order & Payment Services



Request

```
{
  "orderId": "ORD123",
  "amount": 100.50,
  "currency": "USD"
}
```

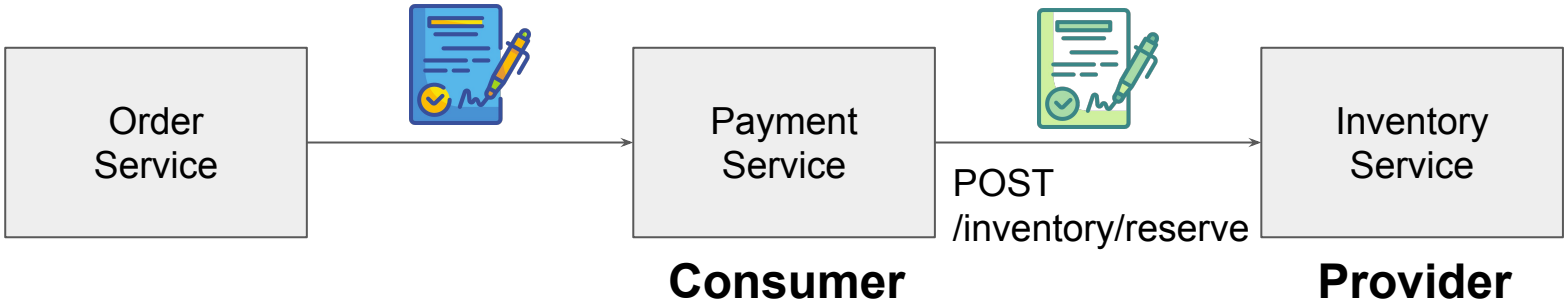
Expected response

```
{
  "paymentId": "PAY987",
  "status": "SUCCESS",
  "orderId": "ORD123"
}
```

Contract between Payment & Order Services

- This contract is stored and published, to be used as a test for the Payment Service
- Running this test requires another mock & contract, this time from Payment to Inventory Service

Contract between Payment & Order Services



Request

```
{
  "orderId": "ORD123",
  "items": [
    { "productId": "P001", "quantity": 2 },
    { "productId": "P002", "quantity": 1 }
  ]
}
```

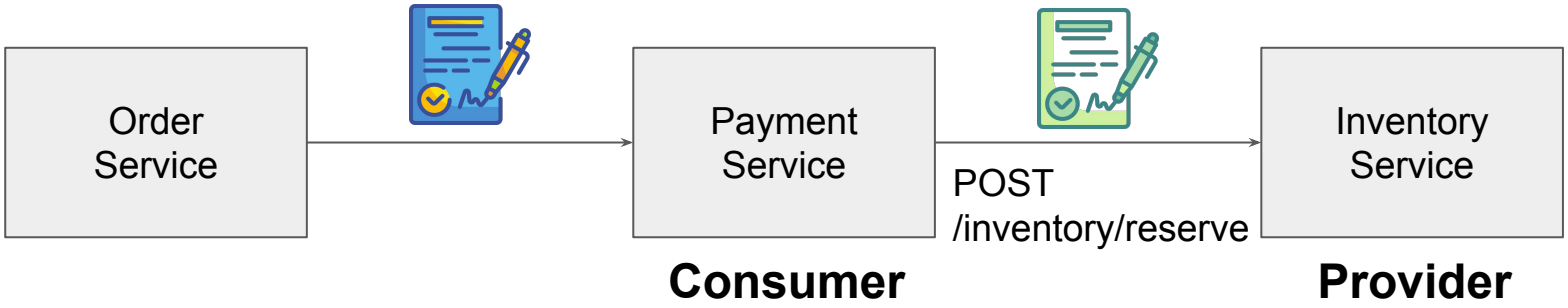
Expected response

```
{
  "orderId": "ORD123",
  "status": "RESERVED"
}
```

Contract between Payment & Order Services

- This second contract is also stored and published, to be used as a test for the Inventory Service
- This whole process can be repeated for other tests for Order Service, resulting in additional contracts

Contract between Payment & Order Services



Request

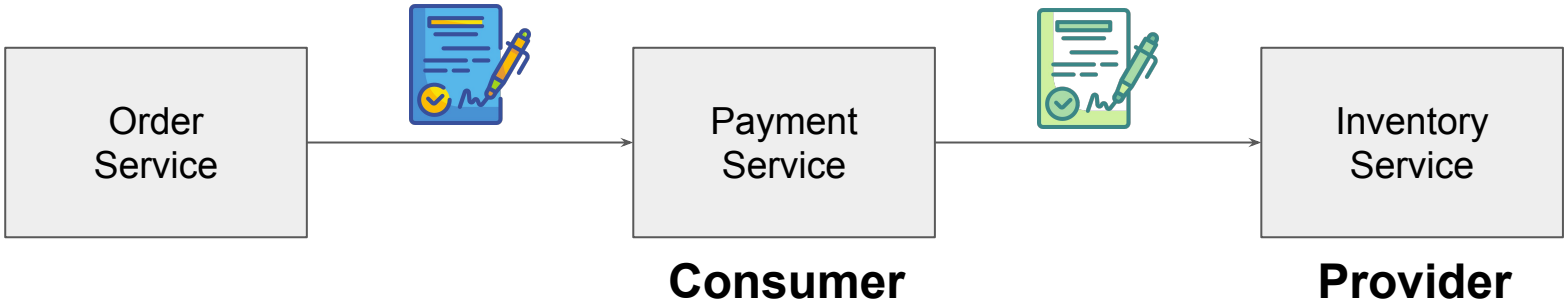
```
{
  "orderId": "ORD123",
  "items": [
    { "productId": "P001", "quantity": 2 },
    { "productId": "P002", "quantity": 1 }
  ]
}
```

Expected response

```
{
  "orderId": "ORD123",
  "status": "RESERVED"
}
```

Note: Running this test requires Inventory Service to have sufficient stock for the items!

Contract with Precondition over Provider State



- In general, a contract may depend on the provider being under a particular state
- Such a contract must also specify **a precondition over the provider state**
  - e.g., “Inventory Service has a sufficient number of items for the reservation to succeed”

Provider Contract Example: Inventory Service

Description: "Reserve all requested items after payment succeeds"

Request: {

```
  "method": "POST",
  "path": "/inventory/reserve",
  "body": { "orderId": "ORD123",
            "items": [ { "productId": "P001", "quantity": 2 }, { "productId": "P002", "quantity": 1 } ]
   }
}
```

ExpectedResponse: {

```
  "status": 200,
  "body": { "orderId": "ORD123",  "status": "RESERVED" }
}
```

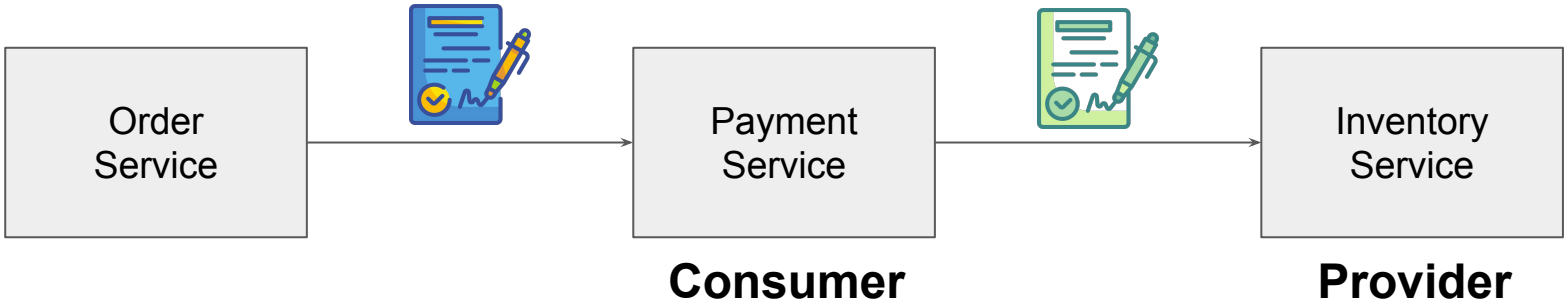
Preconditions: [

```
  "Inventory Service is reachable and accepting requests.",
  "Product P001 exists and has available stock of at least 2 units.",
  "Product P002 exists and has available stock of at least 1 unit.",
  "No reservation has already been completed for order ORD123."
]
```

Create contracts for edge/failure cases, too!

Description: "Return an out-of-stock result when requested quantity exceeds available inventory"
Request: {
  "method": "POST",
  "path": "/inventory/reserve",
  "body": { "orderId": "ORD125",
              "items": [{ "productId": "P001", "quantity": 10 }]
   }
}
ExpectedResponse: {
  "status": 409,
  "body": { "orderId": "ORD125", "status": "REJECTED", "reason": "INSUFFICIENT_STOCK" }
}
Preconditions: [
  "Product P001 exists.",
  "Available stock for P001 is less than 10 units.",
  "No reservation has already been completed for order ORD125."
]

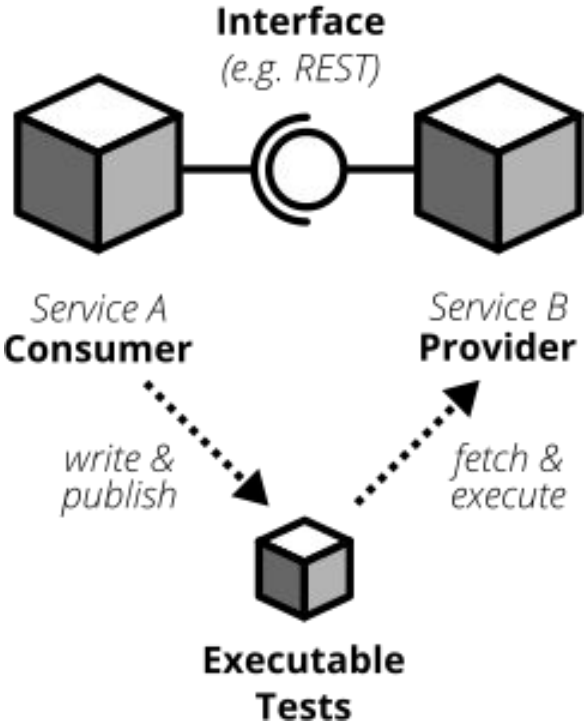
Contract with Precondition over Provider State



- In general, a contract may depend on the provider being under a particular state
- Such a contract must also specify **a precondition over the provider state**
- When testing the provider, the developer writes setup code to bring the provider into a state that satisfies this precondition

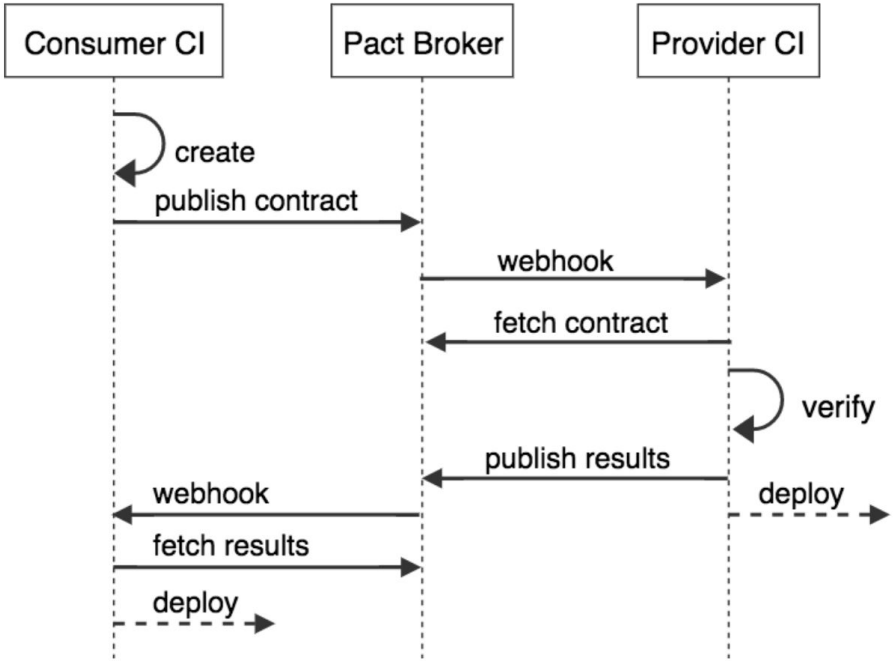
Contract Testing: Benefits

- Allows services to be tested without having to run all of them
- When a provider changes, contracts can be used as regression tests, to detect whether the change affects its consumers
- **Q. How are contracts here different from interface specifications?**



Activity: Write Contracts for Project Services

- Link to the shared Google Doc
- Work with your team members
- For your scheduling application or the service that you are developing for M3/M4:
  - Develop a unit test to test behavior that involves a dependency on an external service
  - Write a contract for the external service, including (1) consumer input, (2) expected provider output, and (3) precondition on the provider state (if necessary)
  - Share the contract with the team for the provider service

Contract Testing Tools

- Pact: An open-source framework for contract testing
- Automates the process of creating, storing, and publishing contracts (Pact Broker)
- But contract testing can be done without a tool!
  - Document & share contracts with other teams!